

Exploring alternative formulations of Smoothed Particle Hydrodynamics for complex fluids and solids

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a Introduction

Complex fluids and solids can be described by the Symmetric Hyperbolic Thermodynamically Compatible (SHTC) equations. These equations have a Hamiltonian structure and are advantageous in problems with large deformations and shock waves. One of the state variables in the SHTC equations is the distortion field (close to the inverse of the deformation gradient). Recently, a new numerical method solving the SHTC equations was developed based on the Smoothed Particle hydrodynamics (SPH) [2].

However, in a later work [1], an alternative definition of the discrete distortion that is useful when adding noise to the SHTC equations (to be applied in continuum physics of nanomaterials) was developed. In this definition, the discrete distortion is defined as the minimizer of the following functional,

$$D_{\mathbf{x}}(\mathbf{A}) = \sum_a \sum_b w_a(\mathbf{x}) w_b(\mathbf{x}) \|\mathbf{X}_{ab} - \mathbf{A}(\mathbf{x}) \mathbf{x}_{ab}\|^2$$

where $\mathbf{X}_{ab}$ is the difference between the Lagrangian positions of the particles a and b , $\mathbf{x}_{ab}$ is the difference between Eulerian positions, and $w_a(\mathbf{x})$ is a weight kernel of particle a (for instance the SPH kernel) and $\mathbf{A}(\mathbf{x})$ is a matrix representation of the distortion field.

This approach is a counterpart to the one used in [2] as we are not evolving the distortion field in time according to the SHTC equations but rather defining it as a minimizer of the functional above at each time step. An advantage of this approach is that in theory it is possible to add fluctuations to the model, which would enable us to simulate nanomembranes and other nanomaterials, where fluctuations play a crucial role.

b Methods

- distortion as an error estimator (compare different definitions)
- comparison with Ondra's SPH

b.1 Distortion as a minimization problem

In this section we will define the distortion field as a minimizer of the following minimization problem:

$$\mathbf{A}(x) = \operatorname{argmin}_{\mathbf{A} \in \mathbb{R}^{d \times d}} \sum_b w(\mathbf{x} - \mathbf{x}_b) \|\mathbf{A}(\mathbf{x} - \mathbf{x}_b) - (\mathbf{X} - \mathbf{X}_b)\|^2 \quad (1)$$

The term

$$e_b(\mathbf{x}) := \|\mathbf{A}(\mathbf{x} - \mathbf{x}_b) - (\mathbf{X} - \mathbf{X}_b)\| \quad (2)$$

plays the role of the error by which we fail to approximate the linearized relation:

$$\mathbf{A}(x) d\mathbf{x} - d\mathbf{X} = 0 \quad (3)$$

The equation 1 can be than written compactly as:

$$\mathbf{A}(x) = \operatorname{argmin}_{\mathbf{A} \in \mathbb{R}^{d \times d}} \sum_b w(\mathbf{x} - \mathbf{x}_b) e_b(\mathbf{x})^2 \quad (4)$$

which is a weighted least squares problem.

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a = \operatorname{argmin}_{\mathbf{A} \in \mathbb{R}^{d \times d}} \sum_b w_{ab} \|\mathbf{A} \mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2 \quad (5)$$

Next we will prove that 1 has a solution:

Lemma: (LS problem) 1 *The LS problem 1 has a solution.*

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Proof:

Let $I(\mathbf{A}_a) := \frac{1}{2} \sum_b w_{ab} \|\mathbf{A}_a \mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2$ and observe that $I(\mathbf{A}_a)$ is a differentiable and coercive function meaning:

$$I(\mathbf{A}_a)/\|\mathbf{A}_a\| \rightarrow +\infty \quad (6)$$

This implies existence of global minimum.

We can also compute this minimum explicitly:

Lets denote $\mathbf{A}_a \mathbf{x}_{ab} - \mathbf{X}_{ab} =: e_{ab}$ and lets take the Gatteaux derivative of $I(\mathbf{A}_a)$ in the direction $d\mathbf{A}_a$

$$\begin{aligned} DI(\mathbf{A}_a)[d\mathbf{A}_a] &:= \lim_{\lambda \rightarrow 0+} \frac{I(\mathbf{A}_a + \lambda d\mathbf{A}_a) - I(\mathbf{A}_a)}{\lambda} = \\ &\lim_{\lambda \rightarrow 0+} \sum_b w_{ab} [(e_{ab} + \lambda d\mathbf{A}_a \mathbf{x}_{ab}, e_{ab} + \lambda d\mathbf{A}_a \mathbf{x}_{ab}) - (e_{ab}, e_{ab})] = \\ &\lim_{\lambda \rightarrow 0+} \frac{1}{2\lambda} \sum_b w_{ab} (2(e_{ab}, \lambda d\mathbf{A}_a \mathbf{x}_{ab}) + \lambda^2 (d\mathbf{A}_a \mathbf{x}_{ab}, d\mathbf{A}_a \mathbf{x}_{ab})) = \\ &\sum_b w_{ab} (\mathbf{A}_a \mathbf{x}_{ab} - \mathbf{X}_{ab}) \cdot d\mathbf{A}_a \mathbf{x}_{ab} = \\ &d\mathbf{A}_a : \left(\sum_b w_{ab} (\mathbf{A}_a \mathbf{x}_{ab} - \mathbf{X}_{ab}) \otimes \mathbf{x}_{ab} \right) \end{aligned} \quad (7)$$

Which immediately implies:

$$\mathbf{A}_a = \mathbf{P}_a \mathbf{Q}_a^{-1} \quad (8)$$

where

$$\mathbf{P}_a = \left(\sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \right) \text{ and } \mathbf{Q}_a = \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right) \quad (9)$$

We can also generalize this and use the definition from Pavelka & Hutter [1] where we hope to achieve better precision at the cost of summing over all particle pairs instead of singletons.

$$\mathbf{A}^{double}(x) = \operatorname{argmin}_{\mathbf{A} \in \mathbb{R}^{d \times d}} \sum_b \sum_c w(\mathbf{x} - \mathbf{x}_b) w(\mathbf{x} - \mathbf{x}_c) \|\mathbf{A} \mathbf{x}_{bc} - \mathbf{X}_{bc}\|^2 \quad (10)$$

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a^{double} = \operatorname{argmin}_{\mathbf{A} \in \mathbb{R}^{d \times d}} \sum_b \sum_c w_{ab} w_{ac} \|\mathbf{A} \mathbf{x}_{bc} - \mathbf{X}_{bc}\|^2 \quad (11)$$

Following the same procedure as in the first example we get the explicit formula for $\mathbf{A}^{double}$ in this case:

$$\mathbf{A}^{double}(\mathbf{x}) = \left(\sum_{a,b} w_a w_b \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \right) \left(\sum_{a,b} w_a w_b \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right)^{-1} \quad (12)$$

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a^{double} = \mathbf{P}_a^{double} (\mathbf{Q}_a^{double})^{-1} \quad (13)$$

where

$$\mathbf{P}_a^{double} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc} \right) \text{ and } \mathbf{Q}_a^{double} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) \quad (14)$$

We can make the following interesting observation that allows us to rewrite $\mathbf{A}^{double}_a$ in an alternative way:

Notice:

$$\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} = (\mathbf{x}_b - \mathbf{x}_c) \otimes (\mathbf{x}_b - \mathbf{x}_c) = \mathbf{x}_b \otimes \mathbf{x}_b + \mathbf{x}_c \otimes \mathbf{x}_c - \mathbf{x}_b \otimes \mathbf{x}_c - \mathbf{x}_c \otimes \mathbf{x}_b \quad (15)$$

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Therefore we have:

$$\begin{aligned} \mathbf{Q}_a^{double} &= \sum_{b,c} w_{ab} w_{ac} (\mathbf{x}_b \otimes \mathbf{x}_b + \mathbf{x}_c \otimes \mathbf{x}_c - \mathbf{x}_b \otimes \mathbf{x}_c - \mathbf{x}_c \otimes \mathbf{x}_b) = \\ &2 \left(\sum_b w_{ab} \right) \left(\sum_b w_{ab} \mathbf{x}_b \otimes \mathbf{x}_b \right) - 2 \left(\sum_b w_{ab} \mathbf{x}_b \right) \otimes \left(\sum_b w_{ab} \mathbf{x}_b \right) \end{aligned} \quad (16)$$

which after introducing the notation: $\bar{\mathbf{x}}_a := (\sum_b w_{ab} \mathbf{x}_b) / (\sum_b w_{ab})$ and realizing that $\sum_b w_{ab} (\mathbf{x}_b - \bar{\mathbf{x}}_a) = 0$ can be written as:

$$\mathbf{Q}_a^{double} = 2 \left(\sum_b w_{ab} \right) \left(\sum_b (\mathbf{x}_b - \bar{\mathbf{x}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right) \quad (17)$$

And similarly:

$$\mathbf{P}_a^{double} = 2 \left(\sum_b w_{ab} \right) \left(\sum_b (\mathbf{X}_b - \bar{\mathbf{X}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right) \quad (18)$$

We can also get rid of the prefactor $2 (\sum_b w_{ab})$ as it will not change the value of $\mathbf{A}_a^{double}$ and simply write:

$$\mathbf{A}_a^{double} = \left(\sum_b (\mathbf{X}_b - \bar{\mathbf{X}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right) \left(\sum_b (\mathbf{x}_b - \bar{\mathbf{x}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right)^{-1} \quad (19)$$

Notice that to compute $\mathbf{A}_a^{double}$ from this definition we only need to know $\bar{\mathbf{x}}_a := (\sum_b w_{ab} \mathbf{x}_b) / (\sum_b w_{ab})$ which can be precomputed.

Next we will make an observation about the order of approximation of $\mathbf{A}_a$ and $\mathbf{A}_a^{double}$.

b.2 Particle distortion field

While it is possible to take the minimizer of the functional $D_{\mathbf{x}}(\mathbf{A})$ directly resulting in the distortion field $\mathbf{A}$ defined as:

$$\mathbf{A}(\mathbf{x}) = \left(\sum_{a,b} w_a w_b \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \right) \left(\sum_{a,b} w_a w_b \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right)^{-1} \quad (20)$$

as suggested in Pavelka et al. [1]. It is worth noticing that in a particle method such as SPH, this definition becomes computationally expensive as it requires to sum over all particle pairs (a, b) in $\text{supp } w_a(\mathbf{x}) \cap \text{supp } w_b(\mathbf{x})$.

To alleviate this issue, we can slightly modify this definition by considering only the diagonal terms of the weight kernels resulting in the following definition of the distortion field $\mathbf{A}_a$ at point $\mathbf{x}_a$ occupied by particle a that is still fully compatible with the SHTC equations under the same assumptions as in [1]:

$$\mathbf{A}_a = \mathbf{P}_a \mathbf{Q}_a^{-1} \quad (21)$$

where

$$\mathbf{P}_a = \left(\sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \right) \text{ and } \mathbf{Q}_a = \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right) \quad (22)$$

The differential of the distortion field with respect to position $\mathbf{x}_a$ is then equal to:

$$d\mathbf{A}_a = d\mathbf{P}_a \mathbf{Q}_a^{-1} + \mathbf{P}_a d\mathbf{Q}_a^{-1} = d\mathbf{P}_a \mathbf{Q}_a^{-1} - \mathbf{P}_a \mathbf{Q}_a^{-1} d\mathbf{Q}_a \mathbf{Q}_a^{-1} \quad (23)$$

Which under the assumptions from [1]:

$$\dot{\mathbf{x}}_a \approx \dot{\mathbf{x}}_b + \nabla \mathbf{v} \Big|_{\mathbf{x}_a} \cdot \mathbf{x}_{ab} \text{ and } \mathbf{X}_{ab} \approx \mathbf{A}_a \mathbf{x}_{ab} \quad (24)$$

simplifies to:

$$d\mathbf{A}_a = -\mathbf{A}_a \nabla \mathbf{v}(\mathbf{x}) \Big|_{\mathbf{x}_a} \quad (25)$$

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giving us the evolution equation for the distortion field which is the evolution equation for the distortion field in the SHTC equations.

Computing $d\mathbf{A}_a$ directly without any approximations yields:

$$d\mathbf{A}_a = -\mathbf{A}_a \mathbf{L}_a + \left(\sum_b \nabla w_{ab} \cdot d\mathbf{x}_{ab} (\mathbf{X}_{ab} - \mathbf{A}_a \mathbf{x}_{ab}) \otimes \mathbf{x}_{ab} \right) \mathbf{Q}_a^{-1} + \left(\sum_b w_{ab} (\mathbf{X}_{ab} - \mathbf{A}_a \mathbf{x}_{ab}) \otimes d\mathbf{x}_{ab} \right) \mathbf{Q}_a^{-1} \quad (26)$$

where:

$$\mathbf{L}_a := \left(\sum_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab} \right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right)^{-1} \quad (27)$$

and it can be seen as an SPH approximation of the velocity gradient at point $\mathbf{x}_a$. If we neglect the terms containing $\mathbf{X}_{ab} - \mathbf{A}_a \mathbf{x}_{ab}$ this reduces to:

$$d\mathbf{A}_a = -\mathbf{A}_a \mathbf{L}_a \quad (28)$$

It remains to see whether this approximation is justified in practice, which we will discuss later.

b.3 Equations of motion

In this section we will derive the equations of motion using equations 28 and 26.

First let $U_h = \sum_b \epsilon_b$ be the total energy of the particle system, let ϵ_b be the specific energy density of particle b . We assume that the specific energy density depends on the density ρ and the distortion field $\mathbf{A}$ ie. $\epsilon_b = \epsilon_b(\rho_b, \mathbf{A}_b)$. Using 28 we can calculate the differential of the energy U_h as:

$$\begin{aligned} dU_h &= \sum_a m_a \epsilon_{\mathbf{A}_a} : d\mathbf{A}_a + \sum_a m_a \epsilon_{\rho_a} d\rho_a = \sum_a m_a \epsilon_{\rho_a} \left(\sum_b m_b d\mathbf{x}_{ab} \cdot \nabla w_{ab} \right) + \sum_a m_a \epsilon_{\mathbf{A}_a} : (-d\mathbf{A}_a) \\ &= \sum_{a,b} m_a m_b \left(\epsilon_{\rho_a} \nabla w_{ab} \cdot d\mathbf{x}_{ab} - \frac{1}{m_b} \epsilon_{\mathbf{A}_a} : \mathbf{A}_a (w_{ab} d\mathbf{x}_{ab} \otimes \mathbf{x}_{ab}) \mathbf{Q}_a^{-1} \right) \\ &= - \sum_{a,b} m_a m_b \left(\frac{1}{m_b} w_{ab} \Pi_a \mathbf{x}_{ab} - \epsilon_{\rho_a} \nabla w_{ab} \right) \cdot d\mathbf{x}_{ab} \\ &= - \sum_{a,b} m_a m_b \left(\frac{1}{m_b} w_{ab} \Pi_a \mathbf{x}_{ab} - \epsilon_{\rho_a} \nabla w_{ab} \right) \cdot (d\mathbf{x}_a - d\mathbf{x}_b) \\ &= - \sum_{a,b} m_a m_b \left(\frac{1}{m_b} w_{ab} \Pi_a \mathbf{x}_{ab} - \epsilon_{\rho_a} \nabla w_{ab} \right) \cdot d\mathbf{x}_a \\ &\quad - \sum_{b,a} m_b m_a \left(\frac{1}{m_a} w_{ba} \Pi_b \mathbf{x}_{ab} - \epsilon_{\rho_b} \nabla w_{ab} \right) \cdot d\mathbf{x}_a \\ &= - \sum_{a,b} m_a m_b \left[w_{ab} \left(\frac{\Pi_a}{m_b} + \frac{\Pi_b}{m_a} \right) \mathbf{x}_{ab} - (\epsilon_{\rho_a} + \epsilon_{\rho_b}) \nabla w_{ab} \right] \cdot d\mathbf{x}_a \end{aligned} \quad (29)$$

where $\Pi_a = \mathbf{A}_a^T \epsilon_{\mathbf{A}_a} \mathbf{Q}_a^{-T}$.

Using Newton's second law $m \dot{\mathbf{v}} = -\frac{dU}{d\mathbf{x}}$ we get the following equations of motion:

$$m_a \dot{\mathbf{v}}_a = \sum_b m_a m_b \left[w_{ab} \left(\frac{\Pi_a}{m_b} + \frac{\Pi_b}{m_a} \right) \mathbf{x}_{ab} - (\epsilon_{\rho_a} + \epsilon_{\rho_b}) \nabla w_{ab} \right] \quad (30)$$

Let us assume that the inner energy ϵ decomposes into bulk and shear part as $\epsilon(\rho, \mathbf{A}) = \epsilon_0(\rho) + \epsilon_s(\mathbf{A})$. We will use the following quadratic relation for the bulk component:

$$\begin{aligned} \epsilon_0(\rho) &= c_0^2/2 \left(\frac{\rho_0}{\rho} - 1 \right)^2 \\ \epsilon'_0(\rho) &= c_0^2 \left(\frac{\rho_0}{\rho} - 1 \right) \left(-\rho_0/\rho^2 \right) = c_0^2 \rho_0 \left(\frac{1}{\rho^2} - \frac{\rho_0}{\rho^3} \right) \end{aligned} \quad (31)$$

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For the shear component we will use the following equation:

$$\begin{aligned}\epsilon_s(\mathbf{A}) &= \frac{c_s^2}{4} \|\text{dev}(\mathbf{A}^T \mathbf{A})\|_F^2 \\ \epsilon'_s(\mathbf{A}) &= c_s^2 \mathbf{A} \text{dev}(\mathbf{A}^T \mathbf{A})\end{aligned}\quad (32)$$

We can than write the Π_a as:

$$\Pi_a = c_s^2 \mathbf{A}_a^T \mathbf{A}_a \text{dev}(\mathbf{A}_a^T \mathbf{A}_a) \mathbf{Q}_a^{-T} \quad (33)$$

We can treat the density ρ in two ways. The first one is the standard SPH density approximation:

$$\rho_a = \sum_b m_b w_{ab} \quad (34)$$

The second way is the following relation:

$$\rho_a = \rho_0 \det(\mathbf{A}_a) \quad (35)$$

With this we have derived the following system of equations for the Particle distortion SPH-SHTC:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \quad (36a)$$

$$\dot{\mathbf{v}}_a = \sum_b m_b \left[w_{ab} \left(\frac{\Pi_a}{m_b} + \frac{\Pi_b}{m_a} \right) \mathbf{x}_{ab} - (\epsilon_{\rho_a} + \epsilon_{\rho_b}) \nabla w_{ab} \right] \quad (36b)$$

$$\mathbf{A}_a = \left(\sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right)^{-1} \quad (36c)$$

The only thing to see now is whether it works or not. We have neglected some terms so here they are (using the notation $\mathbf{e}_{ab} = \mathbf{X}_{ab} - \mathbf{A}_a \mathbf{x}_{ab}$):

$$\sum_a m_a \epsilon_{\mathbf{A}_a} : \left(\sum_b w_{ab} \mathbf{e}_{ab} \otimes d\mathbf{x}_{ab} \right) \mathbf{Q}_a^{-1} = \sum_{a,b} w_{ab} (m_a \Pi'_a \mathbf{e}_{ab} - m_b \Pi'_b \mathbf{e}_{ba}) \cdot d\mathbf{x}_a \quad (37)$$

$$\sum_a m_a \epsilon_{\mathbf{A}_a} : \left(\sum_b \nabla w_{ab} \cdot d\mathbf{x}_{ab} \mathbf{e}_{ab} \otimes \mathbf{x}_{ab} \right) \mathbf{Q}_a^{-1} = \sum_{a,b} ((m_a \Pi'_a \mathbf{e}_{ab} - m_b \Pi'_b \mathbf{e}_{ba}) \cdot \mathbf{x}_{ab}) \nabla w_{ab} \cdot d\mathbf{x}_a \quad (38)$$

where $\Pi'_a = c_s^2 \mathbf{Q}_a^{-1} \text{dev}(\mathbf{A}_a^T \mathbf{A}_a) \mathbf{A}_a^T$ also note that $\Pi_a = (\Pi'_a \mathbf{A}_a)^T$ furthermore we will write $\mathbf{s}_{ab} = (m_a \Pi'_a \mathbf{e}_{ab} - m_b \Pi'_b \mathbf{e}_{ba})$.

The full form of the evolution equation for velocity then reads:

$$\dot{\mathbf{v}}_a = \sum_b m_b \left[w_{ab} \left(\frac{\Pi_a}{m_b} + \frac{\Pi_b}{m_a} \right) \mathbf{x}_{ab} - (\epsilon_{\rho_a} + \epsilon_{\rho_b}) \nabla w_{ab} - \left(w_{ab} \frac{1}{m_a m_b} \mathbf{s}_{ab} + \mathbf{s}_{ab} \cdot \mathbf{x}_{ab} \frac{1}{m_a m_b} \nabla w_{ab} \right) \right] \quad (39)$$

b.3.1 The full evolution equations for $\mathbf{A}$

Here we will try to derive the full evolution equations for the general form of $\mathbf{A}$ from Pavelka and Hutter [1]. The distortion field $\mathbf{A}$ is assumed to have the following full form:

$$\mathbf{A}(\mathbf{x}) = \left(\sum_{a,b} w_a w_b \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \right) \left(\sum_{a,b} w_a w_b \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right)^{-1} \quad (40)$$

b.4 Beryllium plate

Remember the terms we neglected earlier because they were small? They were not that small after all, lets take a look at what happens in the Beryllium plate benchmark without them.

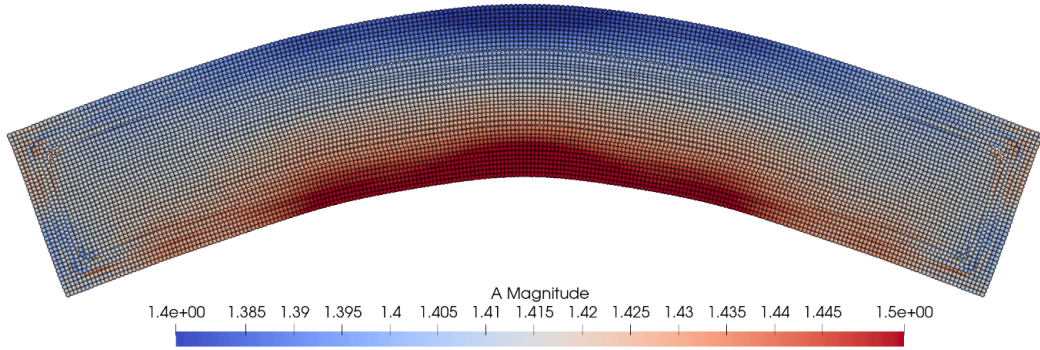


Figure 1: For small deformations and large support radius it seems to work at first.

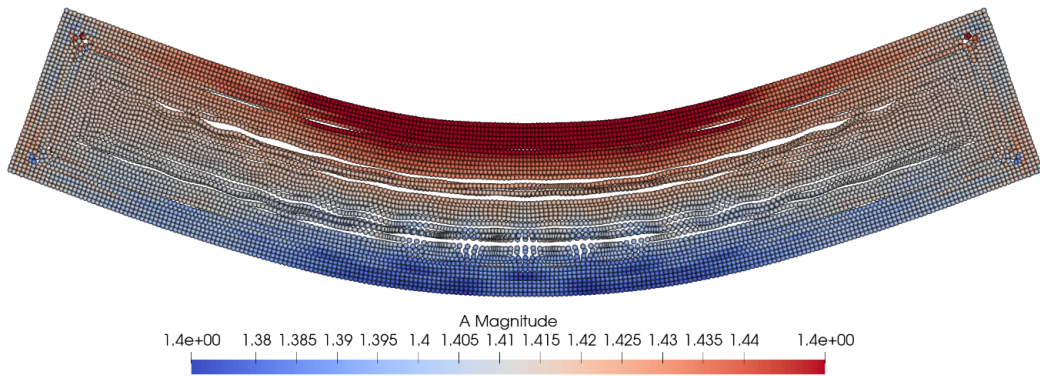


Figure 2: However, even with support radius $h = 20dr$ (which is huge) the simulation becomes unstable after some time. This does not seem to be tensile instability as the pressure is positive in the problematic places. It more has to do with this approximation $\mathbf{X} - \mathbf{A}\mathbf{x} \approx 0$.

TODO: Add the missing terms to the evolution equation for distortion to see what happens.

References

- [1] M. Hütter and M. Pavelka. Particle-based approach to the eulerian distortion field and its dynamics. *Continuum Mechanics and Thermodynamics*, 35(5):1943– 1967, 2023.
- [2] O. Kincl, I. Peshkov, M. Pavelka, and V. Klika. Unified description of fluids and solids in smoothed particle hydrodynamics. *Applied Mathematics and Computation*, 439:127579, 2023.